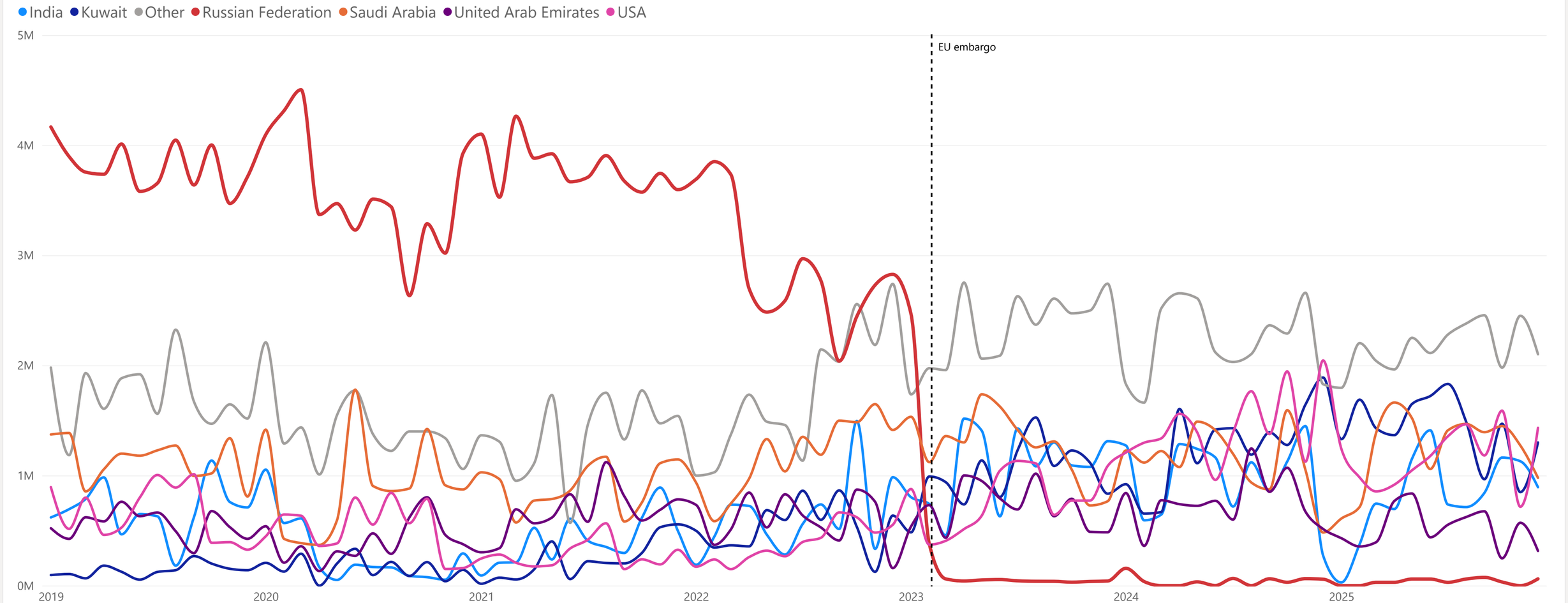
Russia's share of EU diesel imports collapsed within one month of the embargo

Extra-zone diesel imports by origin, monthly, 2019–2025

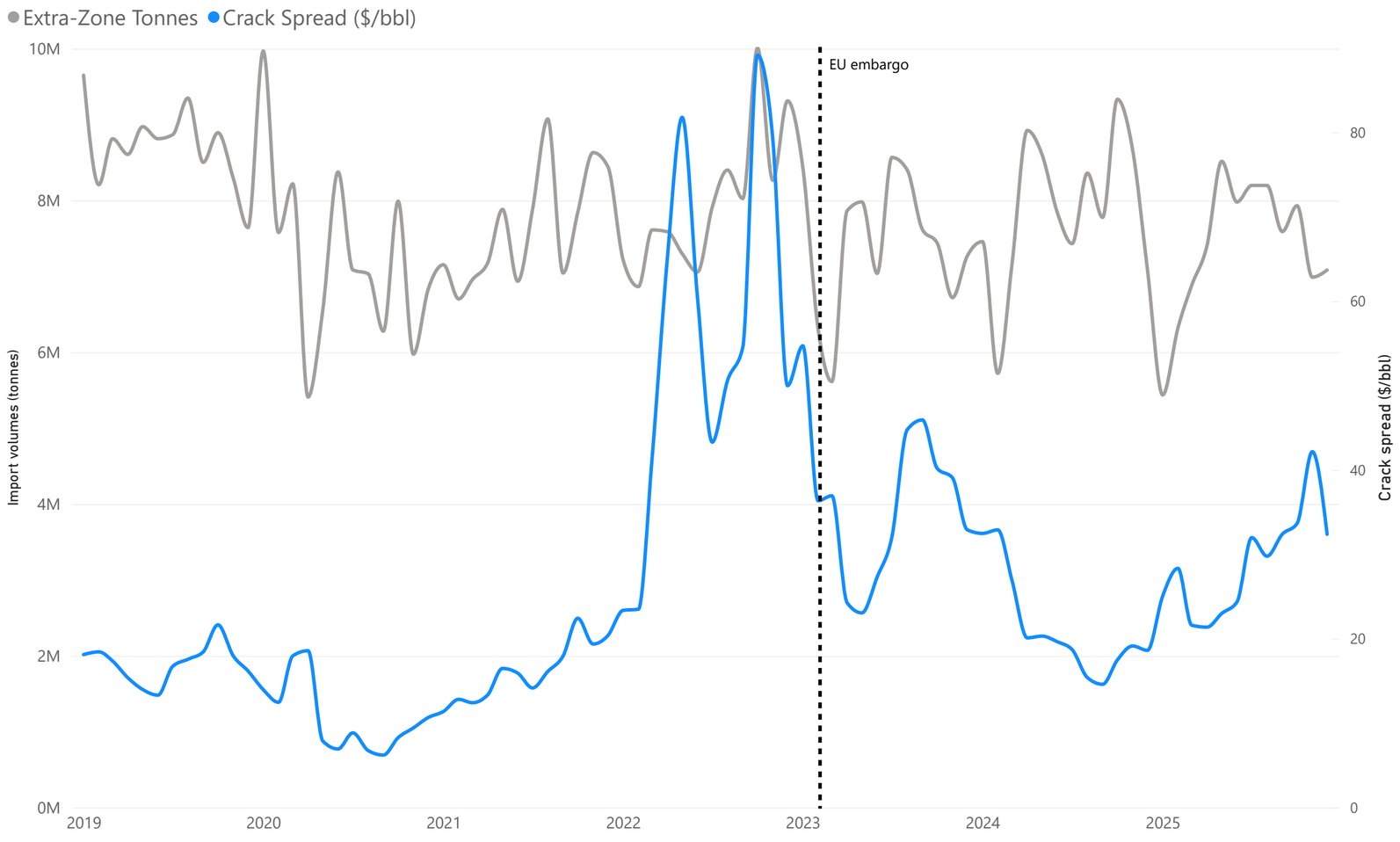
Russia's share before embargo	Time to fall below 2% share	Russia's share after embargo
44.9%	1 month	0.6%



Diesel import volumes did not track the refining margin

Extra-zone import volumes (left axis) vs. crack spread in $/bbl (right axis), monthly, 2019–2025

Max correlation (|r|), any lag	Lags tested	Crack spread peak, Oct 2022 ($/bbl)
0.18	0–6 months	89.2



Why no correlation?

- Tested at lags of 0–6 months, on levels and month-over-month changes
- Max |r| = 0.18, with inconsistent signs across lags
- The crack spread peaked in Oct 2022, months before the embargo
- Reading: the rerouting was driven by the embargo itself, a discrete shock, not by a gradual price response
- Caution: dual-axis charts can suggest co-movement that the statistics do not support

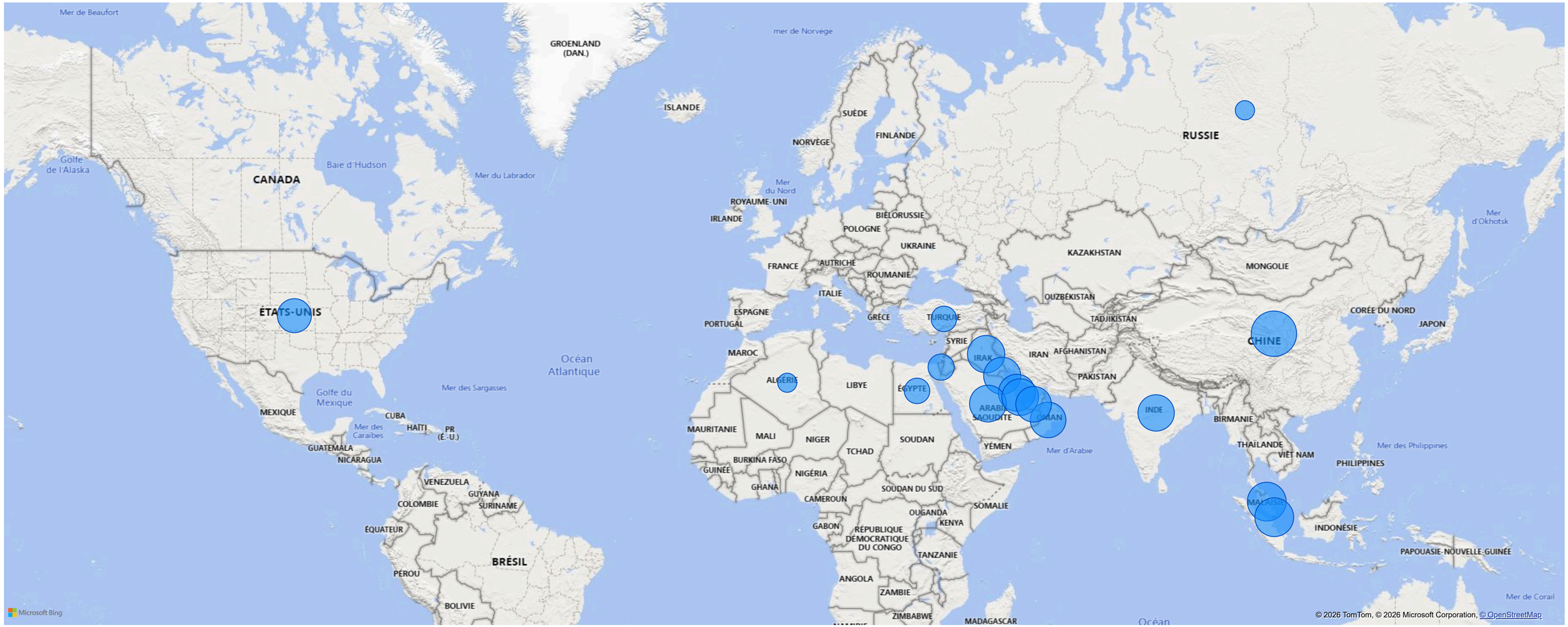
Key dates

- Jun 2022: EU adopts the embargo on Russian refined products
- Oct 2022: crack spread peaks at $89/bbl
- 5 Feb 2023: embargo enters into force
- Mar 2023: Russia's share falls below 2%

EU diesel now travels 53% further on average

Sea distance from each origin's main export port to Rotterdam · 18 countries, 95.6% of extra-zone volume

Increase in shipping demand (tonne-miles)	Average distance before embargo	Average distance after embargo
29.5%	**4509 nm**	**6905 nm**



Methodology & limitations

How the data was built and what it can and cannot tell you

30
reporting countries

2019-2025
84 months of monthly data

240
Comtrade API calls, 0 failures

95.6%
of extra-zone volume with distance data

Data sources

• UN Comtrade: monthly diesel imports declared by 30 countries (EU-27 + UK, Norway, Switzerland)
• US EIA: monthly Brent crude and NY Harbor diesel spot prices
• Sea-route distances to Rotterdam: hand-built reference, 18 origin countries

Key definitions

• Diesel: HS code 271019 (medium distillates), closest available proxy
• Extra-zone: imports whose origin lies outside the 30 reporting countries
• Crack spread: diesel price minus crude price ($/barrel), the refining margin
• Tonne-mile: tonnes shipped × nautical miles travelled, a measure of shipping demand

Pipeline

Comtrade & EIA APIs → Python (cleaning, validation) → DuckDB (SQL analysis) → Power BI
• Fully reproducible from the public GitHub repository
• Import declarations used (Russia stopped publishing exports after 2022)

Data quality fixes

• Double counting removed: transport/customs breakdown rows filtered out
• 14% gap vs. reported totals traced to zero-weight "World" lines (FRA, GBR, ESP)
• Missing zero-flow rows rebuilt, only for months a country actually reported
• Rotterdam/Antwerp transit isolated: 60% / 77% of NL/BE imports are re-exports

Limitations

• Austria excluded (no monthly data for most of the period)
• 2026 excluded (incomplete reporting)
• Diesel price is a US benchmark used as a global proxy
• Distances: one port per country; 6 verified, 1 proxy port, 12 estimated

Go further

• Q2 found no price/volume correlation at any lag
• Full methodology, code and data checks in the GitHub repository

See Q2 details

View code on GitHub